# 🚀 Python Application Deployment on AWS EC2 using Apache & Gunicorn

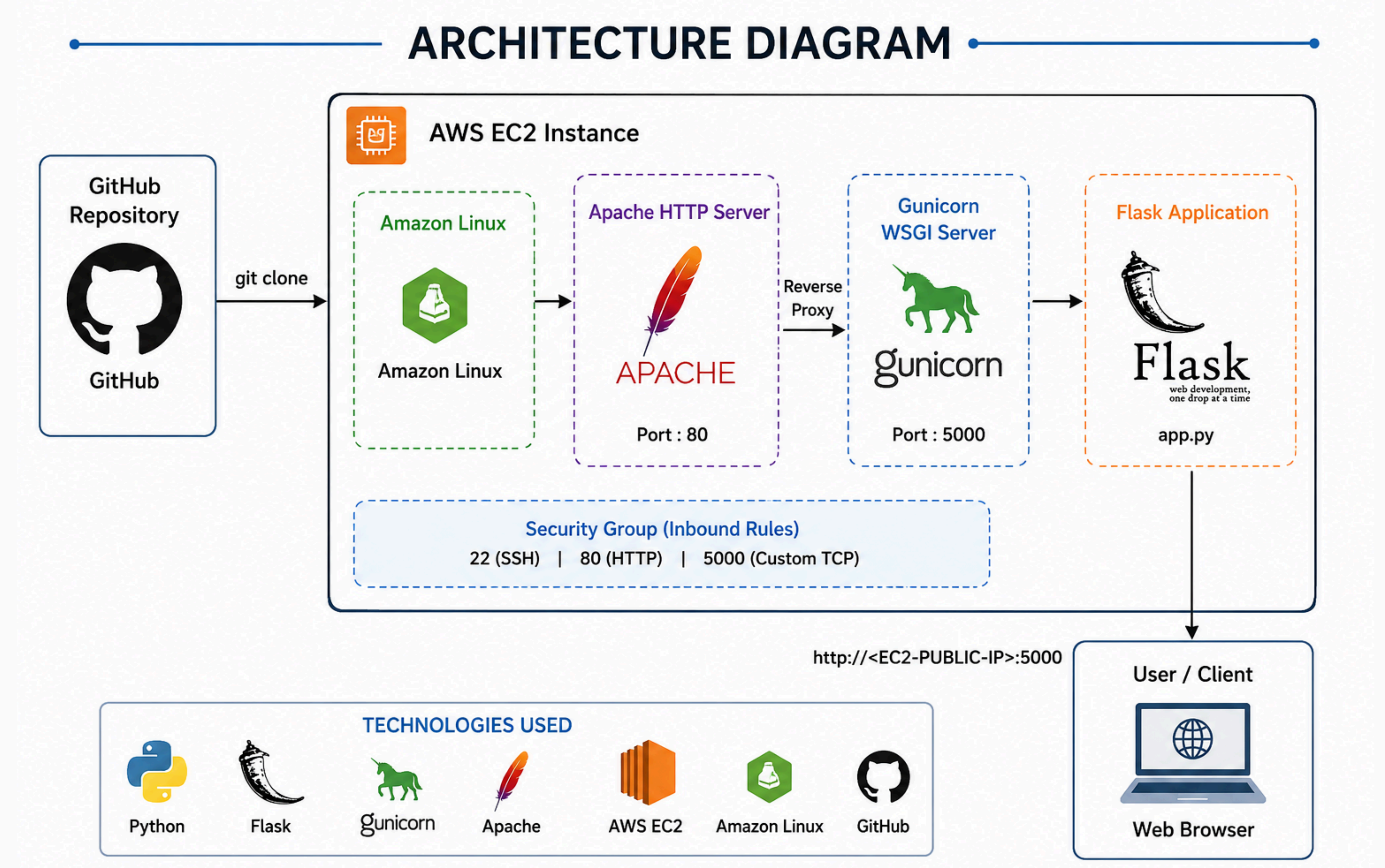
## Project Introduction

**Python-App Deployment** is a web-based application developed using Python, where the application runs using a built-in server and can also be deployed using Gunicorn for production-level performance.

Project objective:

Launching and configuring AWS EC2 instances Hosting Python web applications on Linux servers Managing Python virtual environments Installing and configuring Apache HTTP Server Running Flask applications using Gunicorn WSGI server Configuring security groups and networking Using GitHub for source code management

## 1. Architecture Diagram

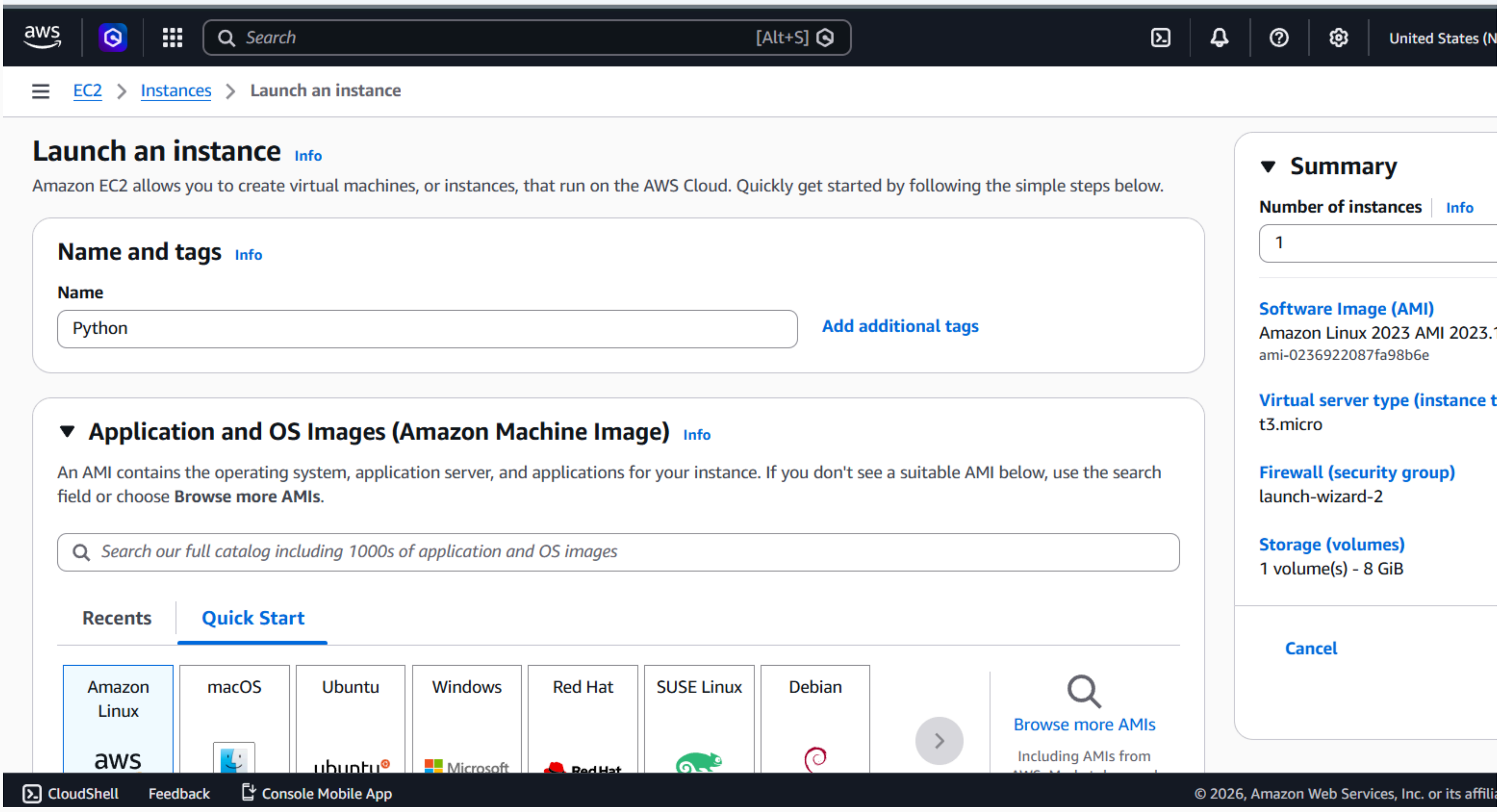



```
⚙️ Technologies Used
🐍 Python 3
🌐 Flask
🦄 Gunicorn
☁️ AWS EC2
🪶 Apache HTTP Server
🐧 Amazon Linux
📂 Git & GitHub
```

```
📁 Project Structure
pythonapp/
│
├── app.py
├── requirements.txt
```

## 2. Instance Setup

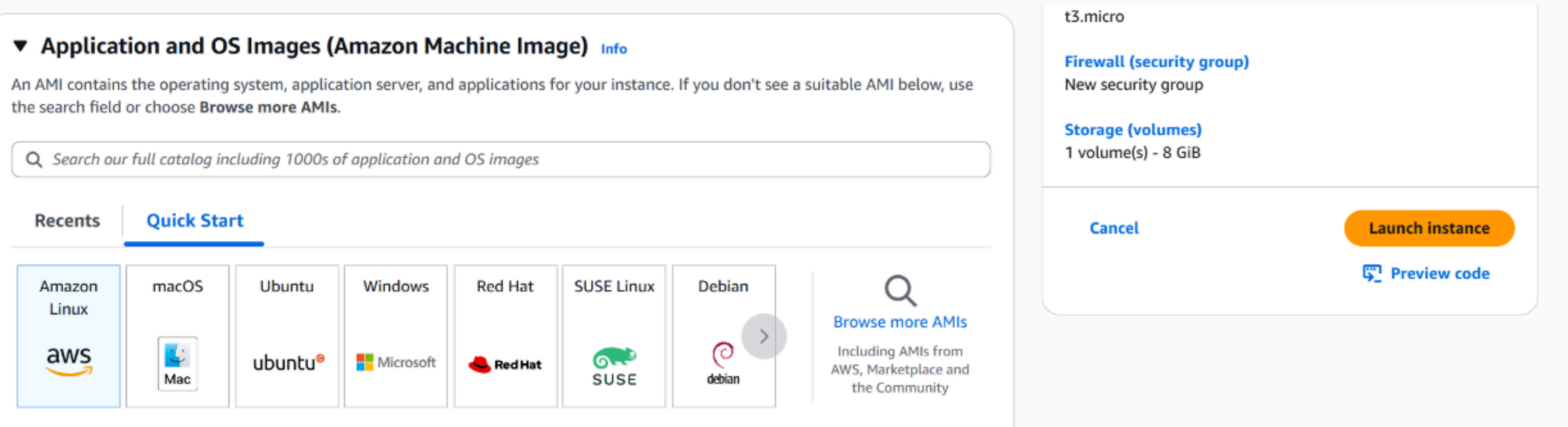
### 2.1 Launch Instance




## 2.2 Names & OS

Name : `Python`

OS : `Amazon Linux`

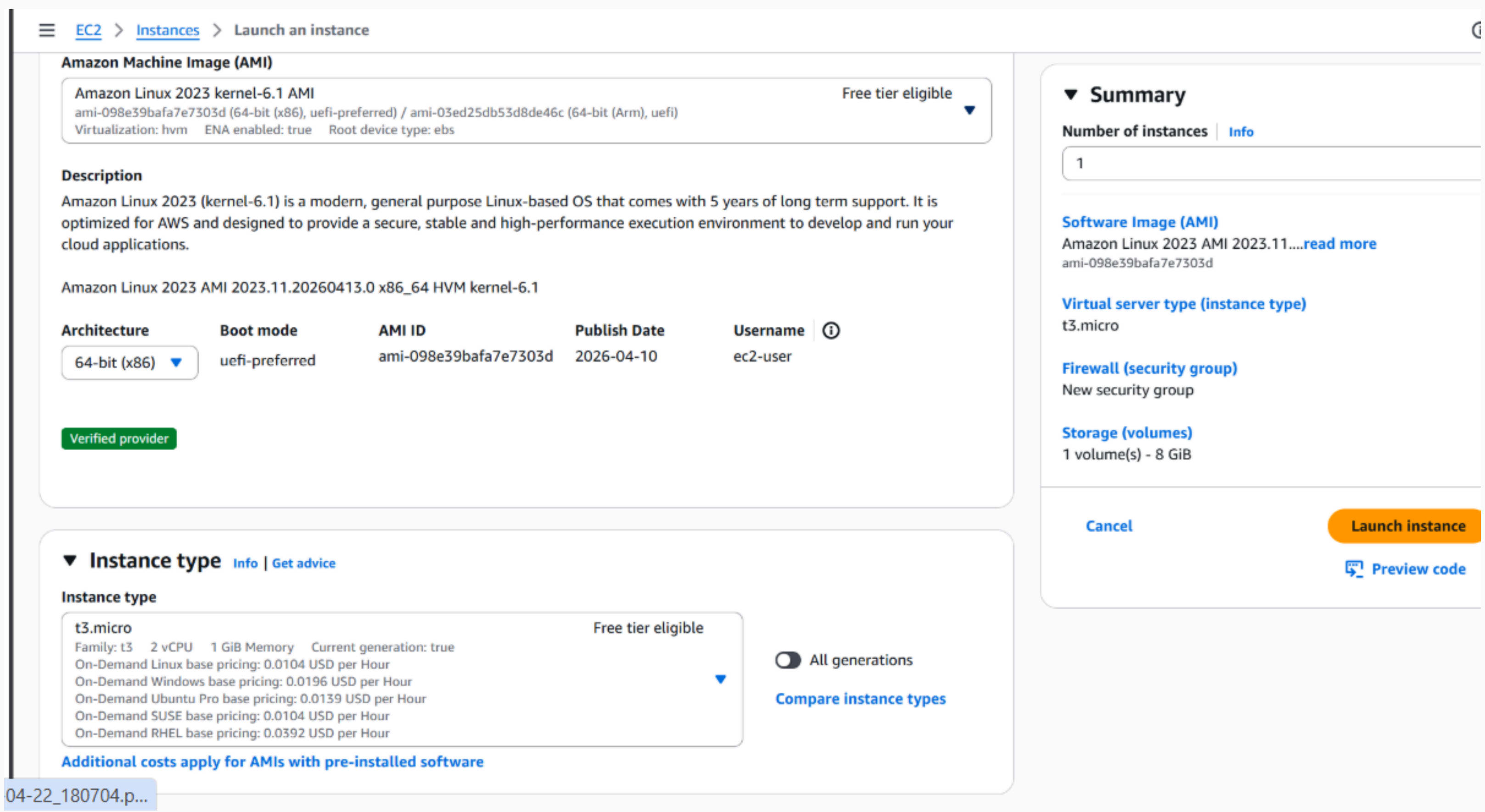



## 2.3 AMI & Instance Type

AMI : `Default`

Architecture : `Default`

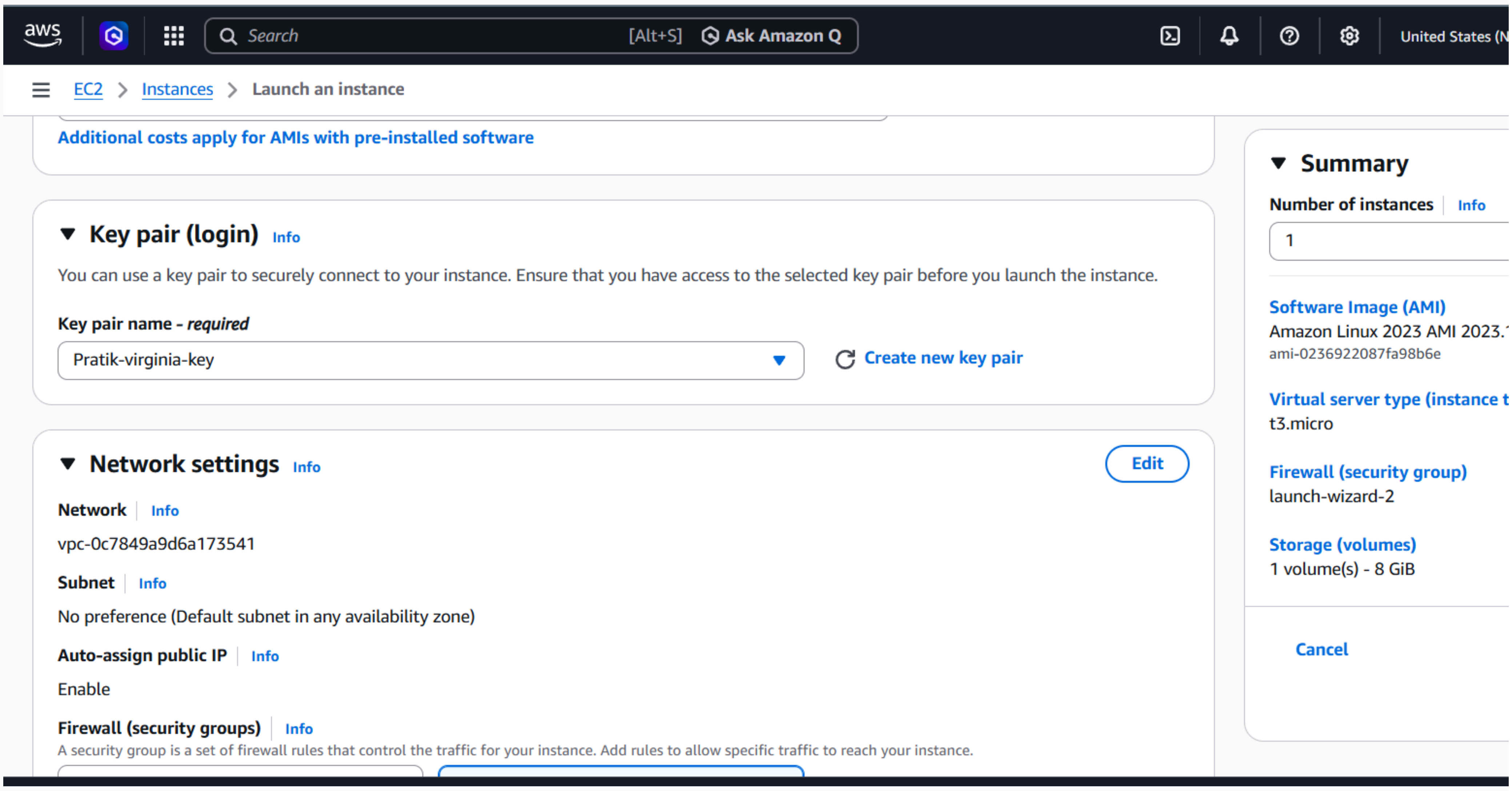
Instance Type : `t3.micro` (2CPU & 1GB Memory)



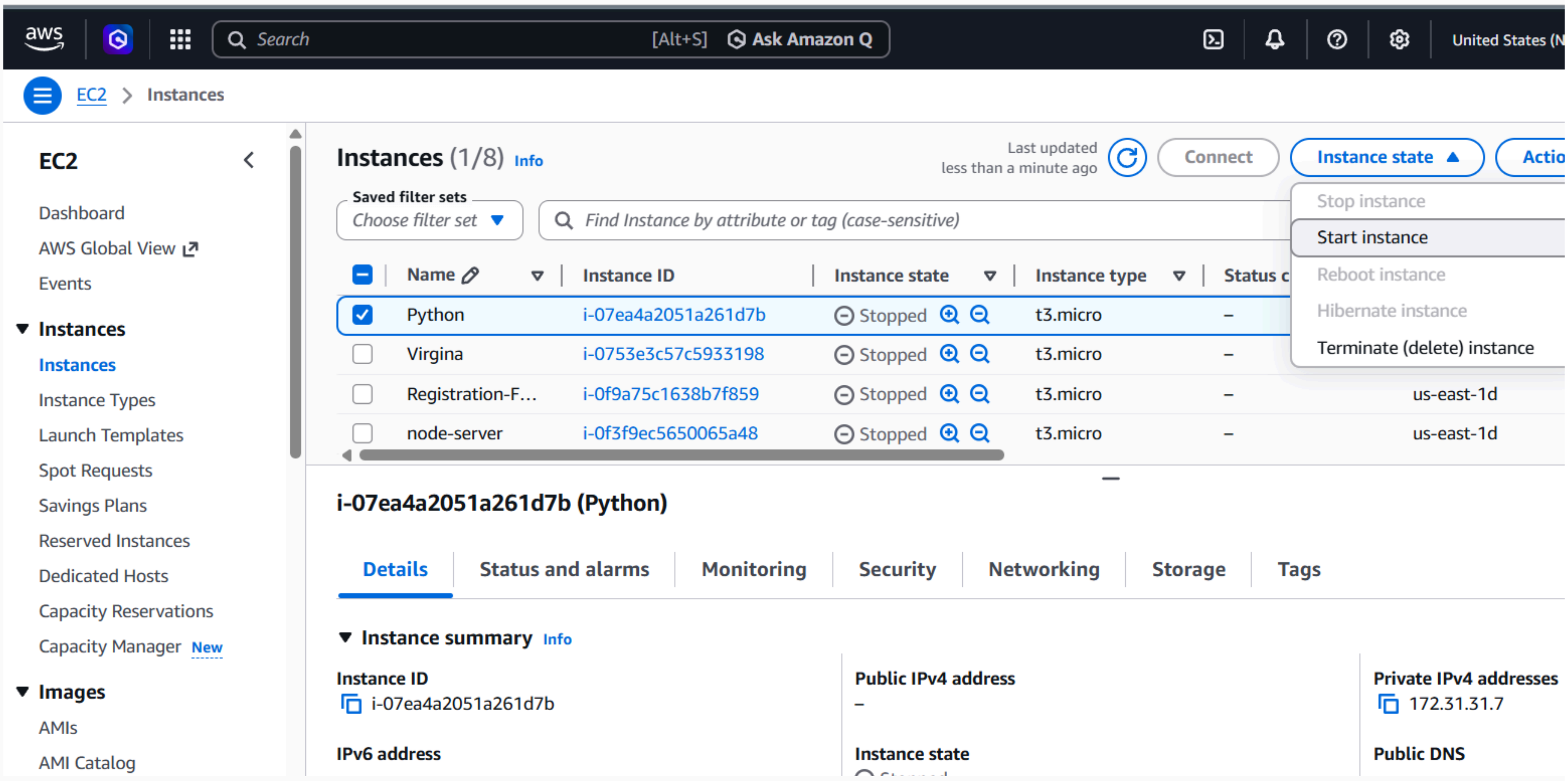

## 2.4 Key & Network

Key : `Pratik-Virginia-Key` (Exiting One)

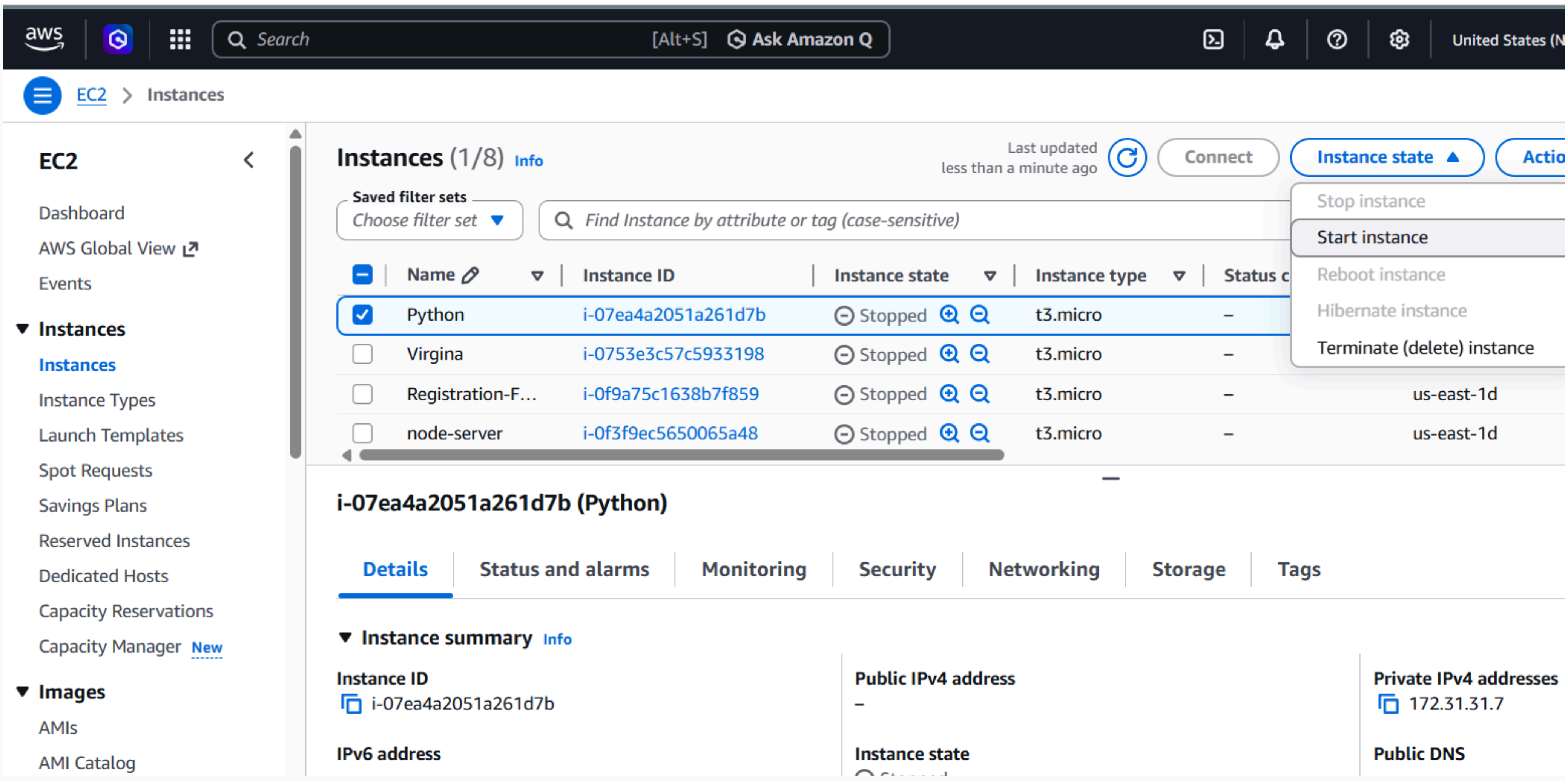
Security Group : `Launch-Wizard-2` (Exiting One)



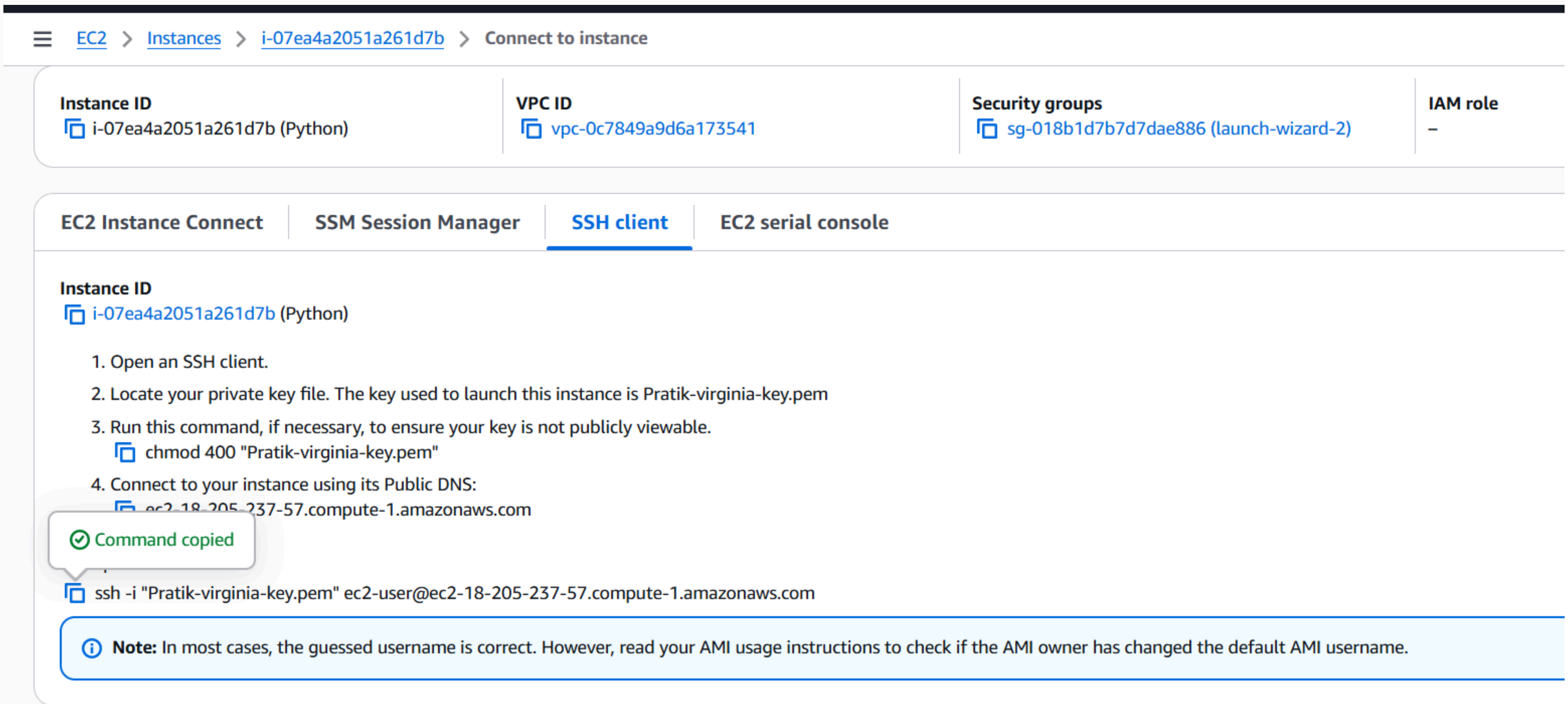

**2.5 Start Instance**




**2.6 Connect**




### 2.7 Copy SSH Client Link




## 3. Installing Packages

### 3.1 Connect Instance With Public IP

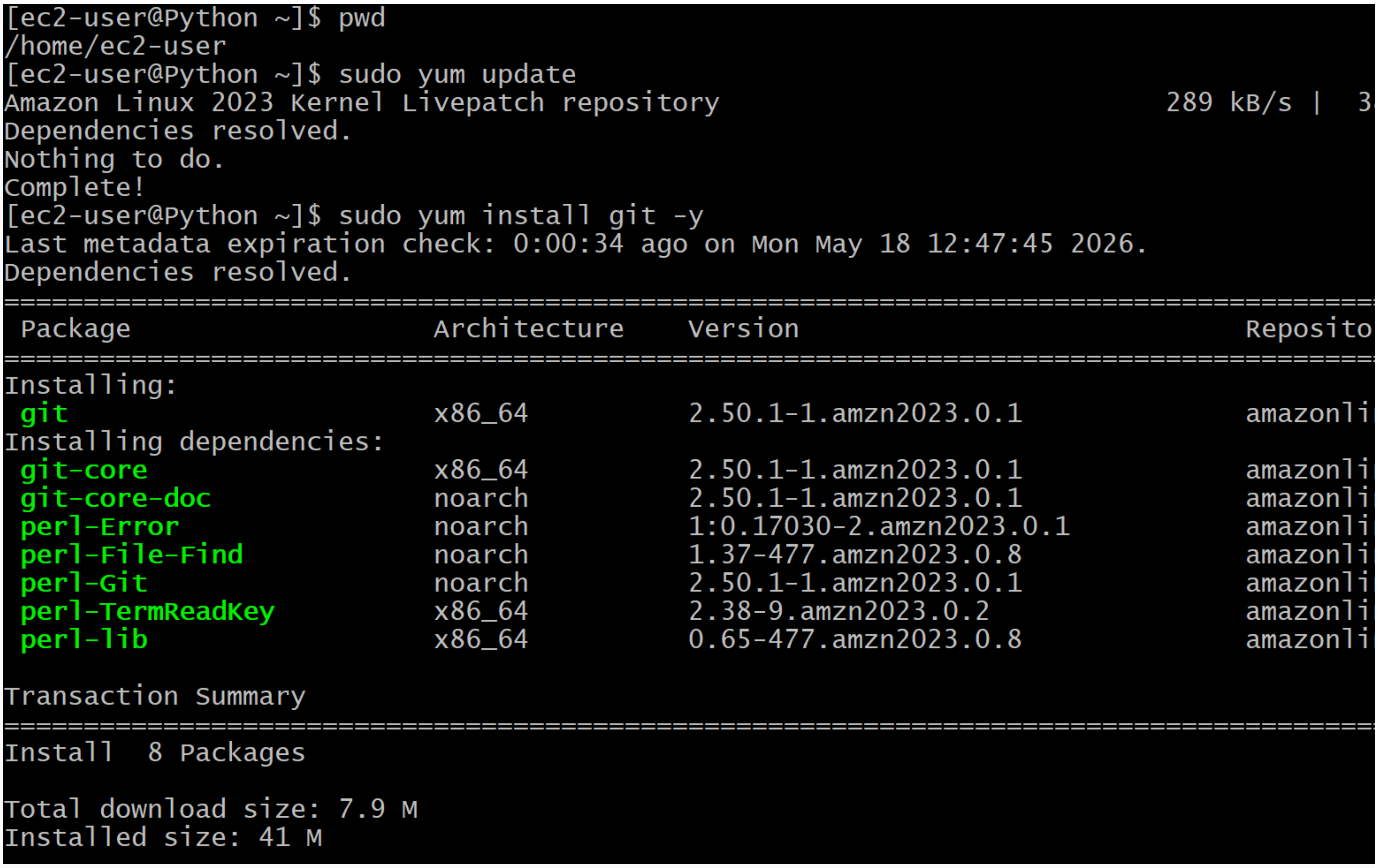
```
ssh -i Pratik-Virginia-Key.pem ec2-user@54.161.38.170
```

### 3.2 Update Packages

```
sudo yum update
```

### 3.3 Install Git

```
sudo yum install git -y
```

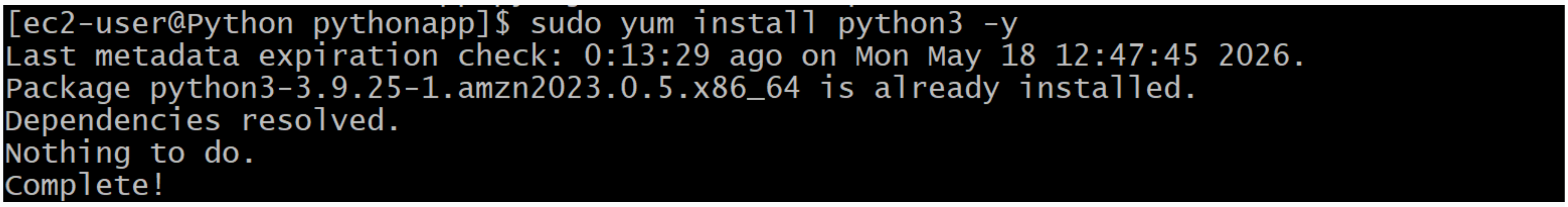



**3.4 Git Clone Repo**

```
git clone https://github.com/iamtruptimane/pythonapp
```

**3.5 Install Python**

```
sudo yum install python3 -y
```

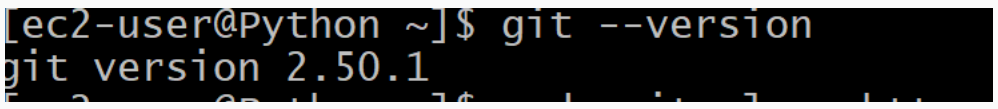



**3.6 Install PIP**

```
sudo yum install python3-pip -y
```

**3.7 View Git, Python & PIP Version**

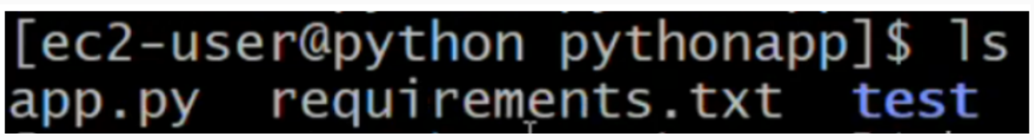
```
git --versio
python --version
pip --version
```




## 4. Running Python App

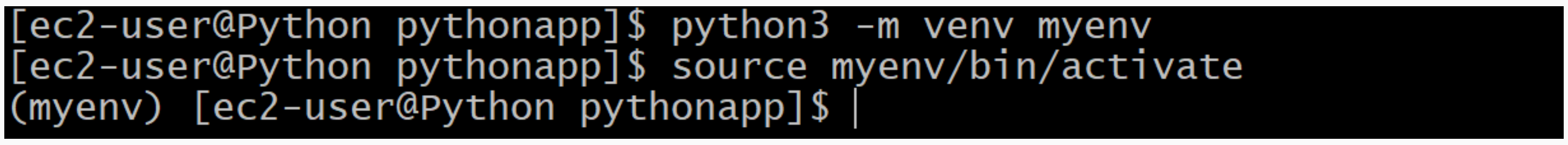
**4.1 Go To Directory Which Create Via Git Clone**

```
cd / pythonapp;
```



#### 4.2 Create Virtual Environment (venv) Inside Project Folder

```
sudo python3 -m venv myenv
```




#### 4.3 Bash Activated venv

```
sudo bash myenv/bin/activate
```

#### 4.4 Install Requirements Inside Project Folder

```
sudo pip install -r requirements.txt
```

#### 4.5 To Run App (FrontGround)

```
python3 app.py

If Press Ctrl + C → App Stops
```

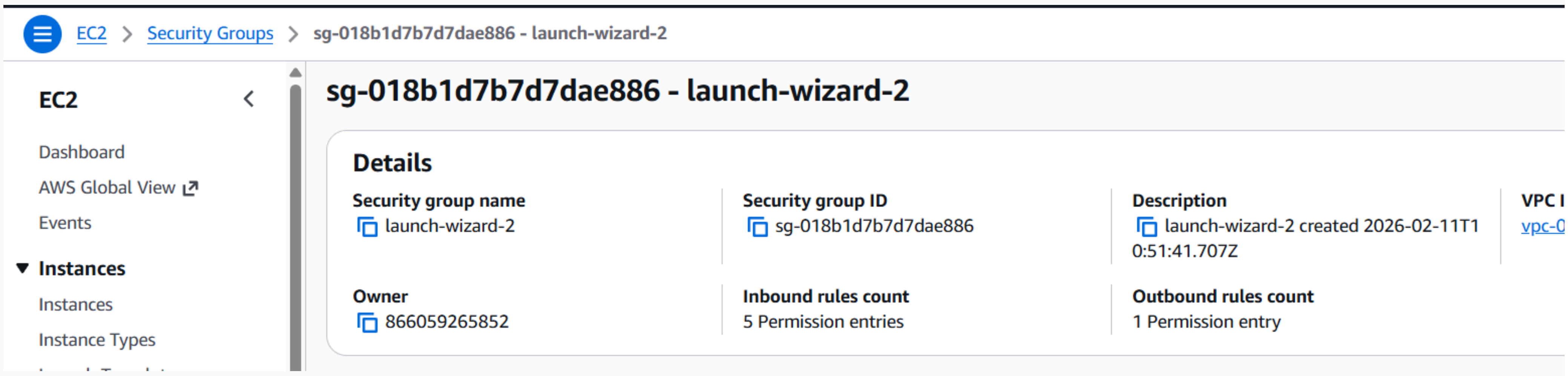
#### 4.6 To Run App In BackGround

```
gunicorn --bind 0.0.0.0:5000 app:app --daemon
```

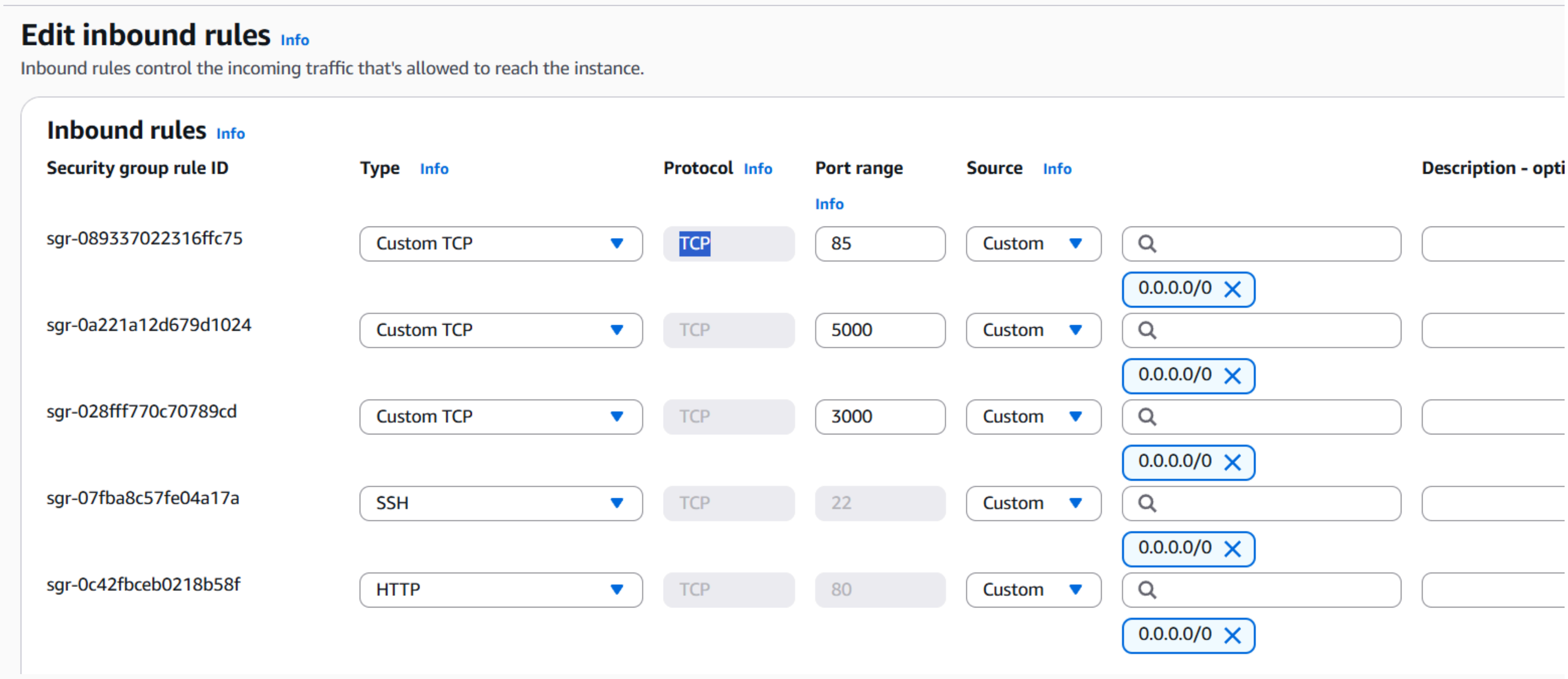
## 5. Access On Browser

#### 5.1 To Add Port Number 5000

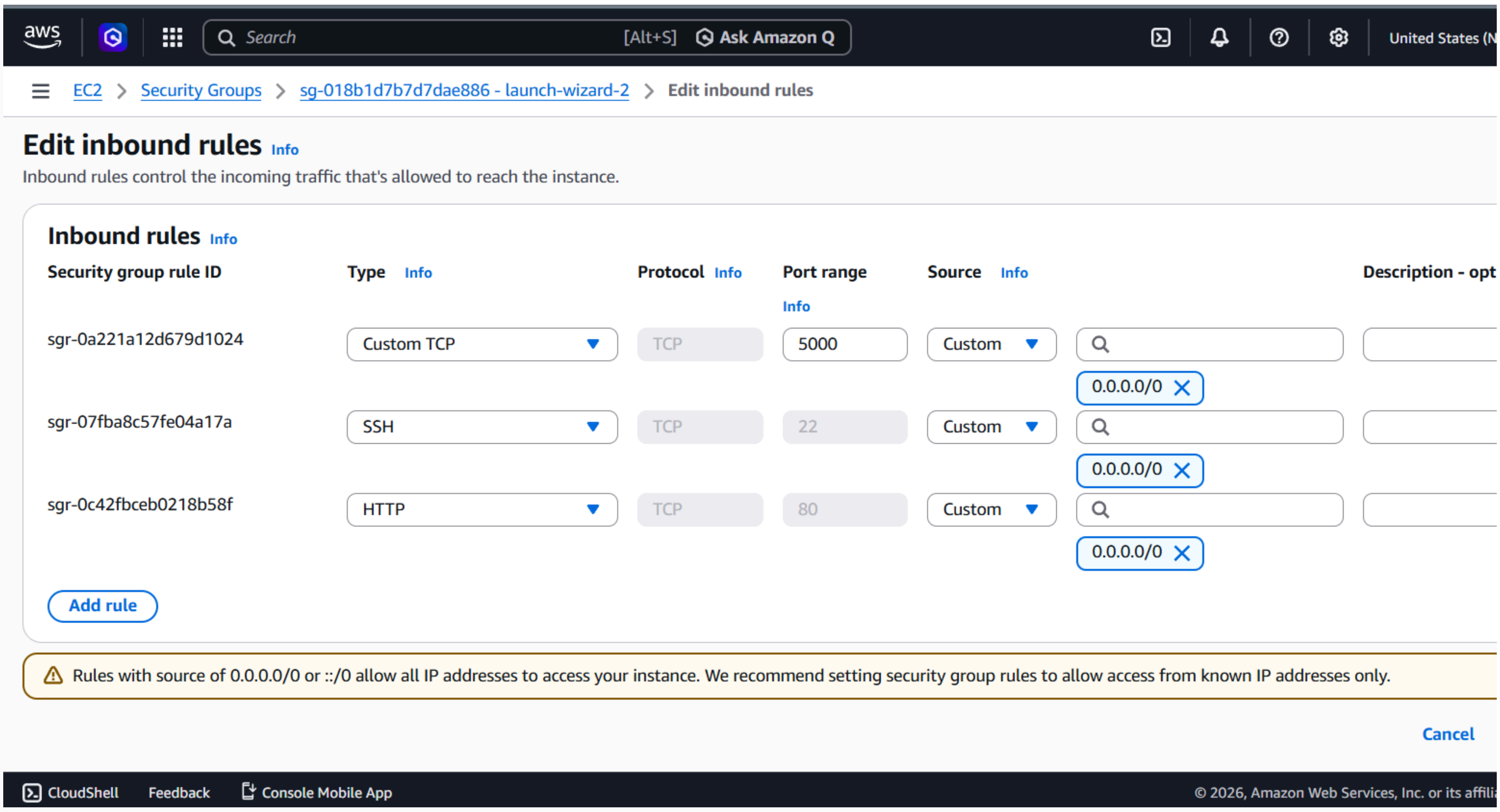
##### 5.1.1 Go To Security Groups




##### 5.1.2 Edit Inbound Rules

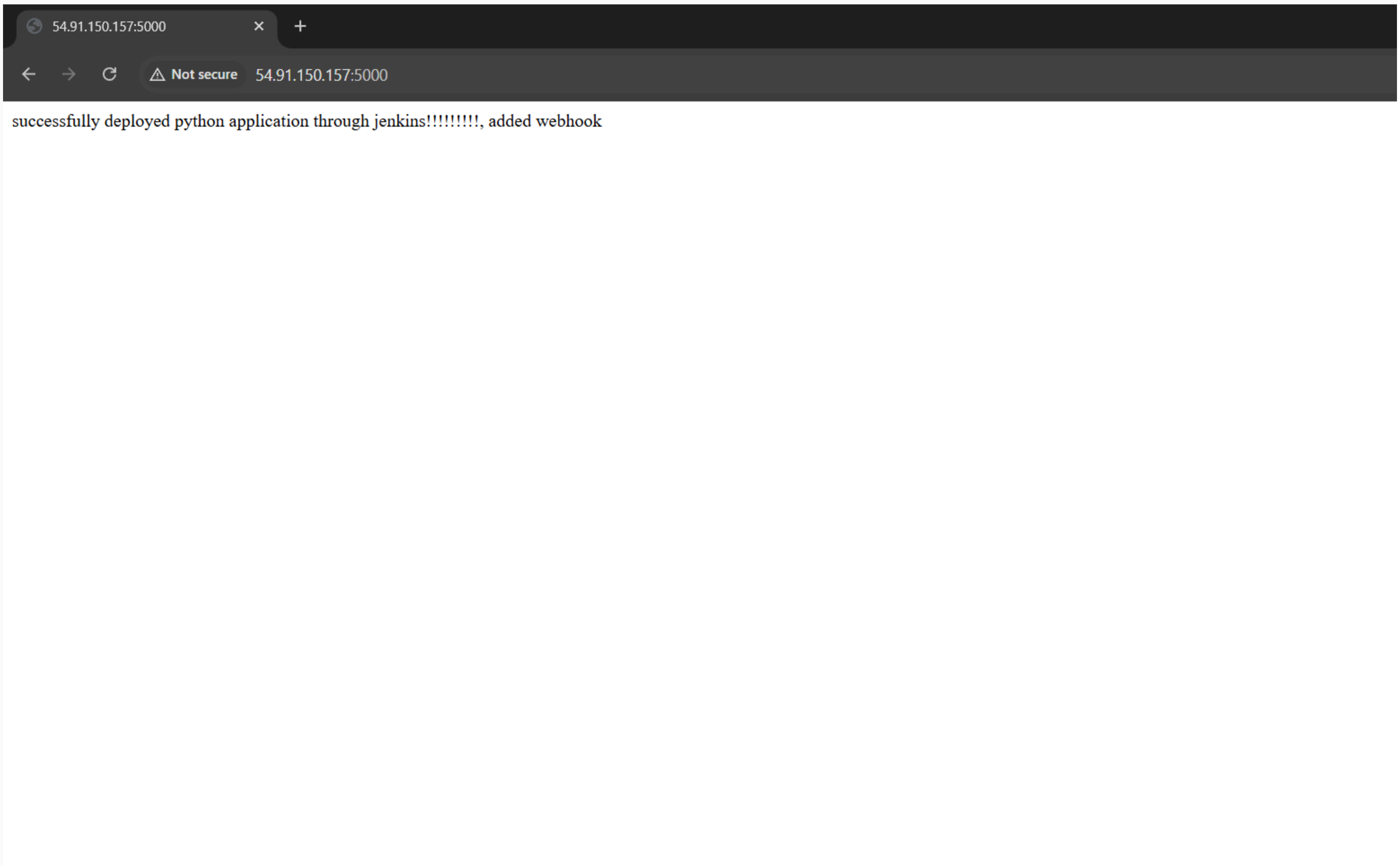



##### 5.1.3 Add Rule → Type : Custom TCP → Port Range : 3000 → Source : Anywhere-IPv4 → Save Rules




#### 5.2 Hit It On Browser And Put File Name

`public ip:5000`




```
# 🔥 Key Learnings

- AWS EC2 setup
- Linux server management
- Python virtual environments
- Flask application deployment
- Gunicorn configuration
- Apache server basics
- GitHub integration
```

## Summary

It Enables Users To Access The Application Through A Public IP On Port 5000, Demonstrating Basic Cloud Deployment And Application Hosting In A Structured Environment.